

Predictive Forecasting of Care Load & Placement Demand

Objective:

The objective of this project was to develop a predictive analytics system to forecast future care load and placement demand using historical intake and discharge data. The system aims to support proactive planning, capacity monitoring, and data-driven decision-making within care management operations.

Problem Statement:

Care facilities face fluctuating intake and discharge volumes, creating uncertainty in resource allocation and staffing. Without forecasting, capacity risks and operational strain cannot be anticipated effectively.

This project addresses:

- Future care load prediction
- System pressure detection
- Capacity breach monitoring
- Model-driven performance comparison

Data and Analytical Approach:

The project utilized historical daily data from the HHS Unaccompanied Alien Children Program, including:

- Intake transfers
- Discharge volumes
- Children currently in care
- CBP custody data

Data preparation included:

- Cleaning and numeric standardization
- Time-series ordering
- Lag and rolling feature engineering
- Net pressure calculation (Intake – Discharge)

Three forecasting models were implemented and compared:

1. Naive Baseline
2. SARIMA (Statistical Model)

3. Random Forest (Machine Learning Model)

Model performance was evaluated using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- MAPE (Mean Absolute Percentage Error)

Key Findings:

- Random Forest achieved the highest predictive accuracy with the lowest MAE, RMSE, and MAPE (~0.25%).
- SARIMA performed significantly better than the baseline but was less accurate than the machine learning model.
- The Naive model confirmed the importance of advanced forecasting techniques.
- Net pressure trends strongly influence care load growth.
- A structural shift observed in early 2025 suggests potential policy or reporting changes impacting flow volumes.

Strategic Recommendations:

- Adopt Random Forest for operational forecasting.
- Monitor net pressure daily to detect system strain early.
- Use forecast outputs for proactive staffing and capacity planning.
- Investigate structural changes in post-2025 data trends.
- Expand system with real-time integration for continuous monitoring.

Business Impact:

The developed dashboard provides:

- Accurate short-term forecasting
- Early-warning capacity indicators
- Clear model comparison for decision confidence
- Data-driven planning support

This solution enhances operational readiness and reduces the risk of unexpected system overload.